ECE 4300 Project

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AES on Microblaze using Nexys A7-100T

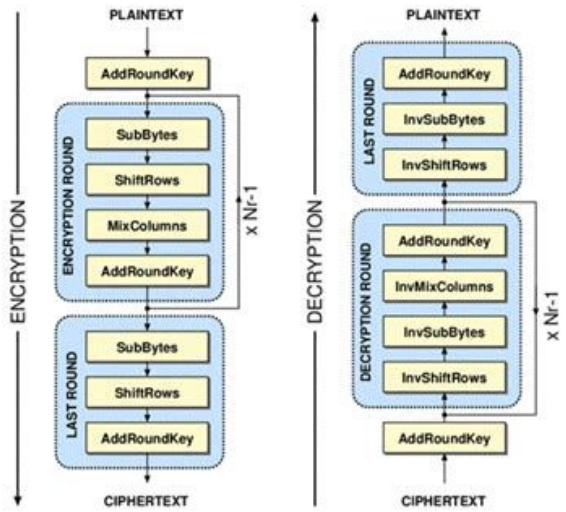

Figure 2: Overall structure of AES algorithm (Source: http://crypto.stackexchange.com/questions/8043/aes-addroundkey).

Setup

- Xilinx FPGA board (Nexys A7-100T)
- Vivado for Hardware programming
- Microblaze softcore processor using Vitis
- Multimeter for power measurements

Benchmarking of AES for 128, 192, and 256-bit key lengths

- Execution Latency (Clock Cycles)
- Cycles per Byte (CPB)
- Throughput (Mbps)
- Power draw (mW)

Gantt Chart

Microblaze AES benchmarking

DATE 2/24/26

WBS NUMBER	TASK TITLE	PCT OF TASK COMPLETE	PHASE ONE			PHASE TWO			PHASE THREE			PHASE FOUR	
			2/23-2/27	3/2-3/6	3/9-3/13	3/16-3/20	3/23-3/27	3/30-4/3	4/6-4/10	4/13-4/17	4/20-4/24	4/27-5/1	5/4-5/8
1	**Initial Setup**												
	Vivado Setup	0%											
	Basic Board tests	0%											
	Vitis Setup	0%											
	Basic Vitis Tests	0%											
2	**Microblaze Implementation**												
	Initial Setup	0%											
	Test programs	0%											
	Test Output	0%											
3	**AES Implementation**												
	Find Implementation/Library	0%											
	Compile	0%											
	Verify Output	0%											
4	**Performance Metrics**												
	Plan Benchmarks	0%											
	Gather Performance Data	0%											
	Compare tests	0%											
	Compile Findings	0%											
5	**Presentation**												
	Buffer Time	0%											
	Finish Reports	0%											
	Prepare Presentation	0%											
	Finished	0%											